

ANTAS GUPTA

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PROFESSIONAL SUMMARY

3+ years of production experience designing ETL pipelines, data warehouses, and BI reporting solutions across Informatica, IBM DataStage, and Microsoft SSIS. Proven track record processing multi-million-row daily loads with embedded data quality validation, reducing batch runtimes by 40%, and delivering Power BI dashboards used daily by stakeholders across multiple units. M.S. Computer Science candidate (AI specialization) with expertise in LLM applications, NLP, and RAG-based retrieval systems. STEM OPT authorized — 3-year work authorization; no employer sponsorship required.

TECHNICAL SKILLS

Languages & Querying: Python, SQL, Java, C, Kotlin, HTML/CSS

ETL & Integration: IBM DataStage, Microsoft SSIS, Informatica, ETL Pipeline Design, Data Quality Validation, Batch Processing, Data Modeling, Dimensional Modeling

Databases: SQL Server, Oracle, MySQL, Database Administration, Indexing, Performance Tuning, Data Warehousing

BI & Visualization: Power BI, Power BI Service, Data Visualization, Dashboard Development, Reporting Automation

Cloud & DevOps: Microsoft Azure, Docker, Linux, Git, GitHub, CI/CD, Azure DevOps, Terraform

ML & AI: PyTorch, TensorFlow, scikit-learn, HuggingFace Transformers, FinBERT, Prompt Engineering, RAG, Embeddings, Semantic Search, NumPy, pandas, matplotlib, Jupyter

Web & APIs: Streamlit, Plotly, BeautifulSoup, REST APIs, yFinance

Security: Post-Quantum Cryptography (Kyber KEM, Dilithium, ChaCha20-Poly1305 AEAD), Secure Messaging, CLI Development

Project Management: JIRA, Agile, Scrum, Continuous Improvement

PROFESSIONAL EXPERIENCE

Secure Messaging Engineering Intern | Platinum Venture Capital Partners (IronHalo Project), New York, NY *Jul 2025 – Sep 2025*

- Built a Python CLI tool with 10+ commands integrating Kyber KEM, Dilithium signatures, and ChaCha20-Poly1305 AEAD into a modular post-quantum cryptographic workflow adopted as the project's core security layer
- Prototyped a forward-secret ratchet mechanism inspired by Signal's Double Ratchet protocol for per-message key rotation, reducing compromise blast radius across distributed messaging systems
- Researched Decentralized Identity (DID) standards and secure messaging workflows, producing a technical memo evaluating enterprise integration pathways

Senior Technical Executive — AI Solutions | SV Technosolutions Pvt. Ltd., New Delhi, India *Jun 2023 – Jul 2024*

- Built and deployed predictive maintenance and anomaly-detection models for Indian Railways infrastructure using Python-based ML workflows, reducing unplanned maintenance events through proactive failure forecasting
- Integrated AI pipelines with operational systems across multiple stakeholder teams; supported end-user training, rollout coordination, and security compliance
- Conducted AI feasibility analysis and operational optimization assessments, translating business requirements into scalable ML solution architectures

Analyst — Data Engineering | Principal Global Services (Principal Financial Group), Pune, India *Aug 2022 – May 2023*

- Designed and maintained 15+ scalable ETL and backend data-processing workflows across Informatica, IBM DataStage, and Microsoft SSIS, integrating enterprise databases and APIs with multi-million-row daily loads
- Developed Power BI dashboards and data quality monitoring reports tracking ETL pipeline performance, freshness SLAs, and validation failures; outputs used daily by data stewards and downstream analysts across 3+ business units
- Collaborated with cross-functional teams in Agile sprints to define data quality acceptance criteria and SLA targets, reducing pipeline-related incidents through proactive monitoring and stakeholder communication

IT Senior Associate — ETL Developer | NTT DATA Services, Gurgaon, India *Aug 2021 – Jul 2022*

- Developed 20+ IBM DataStage ETL workflows for Blue Cross Blue Shield of Rhode Island (BCBSRI), processing HIPAA-governed claims, member, and provider data feeds into downstream reporting systems handling 5M+ records per nightly batch
- Optimized enterprise DataStage workflows via partitioning, parallelism tuning, and debugging, reducing runtimes on critical nightly batch jobs by up to 40% and improving downstream system availability
- Ensured data quality and HIPAA compliance across all pipeline outputs through automated validation checks, audit logging, and incident escalation protocols aligned with client SLA requirements

AI/ML Research Intern | Unique Identification Authority of India (UIDAI), Govt. of India, Bengaluru, India *Apr 2021 – Aug 2021*

- Contributed to Face Authentication Modality and liveness detection models for edge-device deployment in India's national digital identity ecosystem (Aadhaar), supporting real-time inference on resource-constrained hardware
- Optimized facial authentication performance across varied lighting, expressions, and poses; supported lightweight model deployment targeting sub-50ms inference latency on edge devices

Cloud Infrastructure Intern | Bharti Airtel, Gurgaon, India *Jun 2019 – Jul 2019*

- Assisted in implementing cloud-based infrastructure solutions including server deployment and automated data backup workflows, supporting telecom-grade reliability standards
- Collaborated with cross-functional teams on cloud performance optimization and cost-efficiency initiatives; prepared technical documentation supporting operational processes and knowledge transfer

PROJECTS

Financial Red Flag Detector — NLP-Based 10-K Risk Scorer | *Spring 2026*

- Built an NLP-driven financial risk intelligence system processing 10+ SEC 10-K filings daily via the EDGAR API, analyzing Risk Factors and MD&A disclosures to surface corporate distress signals with sentence-level explainability
- Designed a retrieval-based document analysis workflow using transformer embeddings and semantic similarity search, enabling contextual financial risk identification across 100+ pages of SEC filings per company
- Applied FinBERT and NLTK-based text processing for financial sentiment analysis, red-flag keyword detection, and filing comparison; generated composite 0–100 risk scores using sentiment severity, disclosure tone, and boilerplate similarity metrics
- Developed an interactive Streamlit and Plotly dashboard integrating SEC EDGAR, yFinance, and explainable AI visualizations to compare historical filing risks, validate predictions against market reactions, and analyze disclosure trends across multiple public companies

EDUCATION

Binghamton University, State University of New York — Binghamton, NY *August 2026*

M.S. Computer Science, Specialization in Artificial Intelligence | GPA: 3.53/4.00

Relevant Coursework: LLM Foundations & Applications, NLP, Machine Learning, Data Mining, AI, Computer Security, Algorithms, Systems Programming

SVKM's Narsee Monjee Institute of Management Studies (NMIMS) — Mumbai, India *Jul 2024*

M.B.A. in Information Technology and Systems Management

JSS Academy of Technical Education — Noida, India *May 2021*

B.Tech. in Electronics and Communications Engineering

PUBLICATIONS

- Gupta, Antas. "5G Mobile Communication Technology: An Overview." International Research Journal of Engineering and Technology (IRJET), March 2018.

CERTIFICATIONS

- Azure Fundamentals (AZ-900), Microsoft
- HashiCorp Terraform Associate (004)
- AI for Everyone, DeepLearning.AI (Coursera)
- MATLAB Machine Learning Onramp, MathWorks
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- RPA Developer Foundations (9 modules), Automation Anywhere